



From Aggregates to Estimates - population inference from MPD and standard data sources

Session 2.4.4 - Review Activity

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The following slides contain examples official population projections and variants of population estimates for selected districts of Ghana. For each district you can see

- The number of population as counted in the 2021 population census in July 2021 (**turquoise**), published by Ghana Statistical Service (GSS)
- The annual population projection for the years 2022 to 2025 (**turquoise**), also published by GSS
- The monthly population estimates based on a simple method (**red**) that applies monthly relative net population change as derived from available CDR data on subscriber stock and flow data per month (& monthly growth rates derived from the GSS population projections)
- The monthly population estimates and CIs (**blue**) based on demographic accounting model and domain-level weighting of flow data per month (& monthly growth rates derived from the GSS population projections) - an approach that takes into account the varying district-level population coverage of the CDR data, as well as a SIM-to-person correction factor

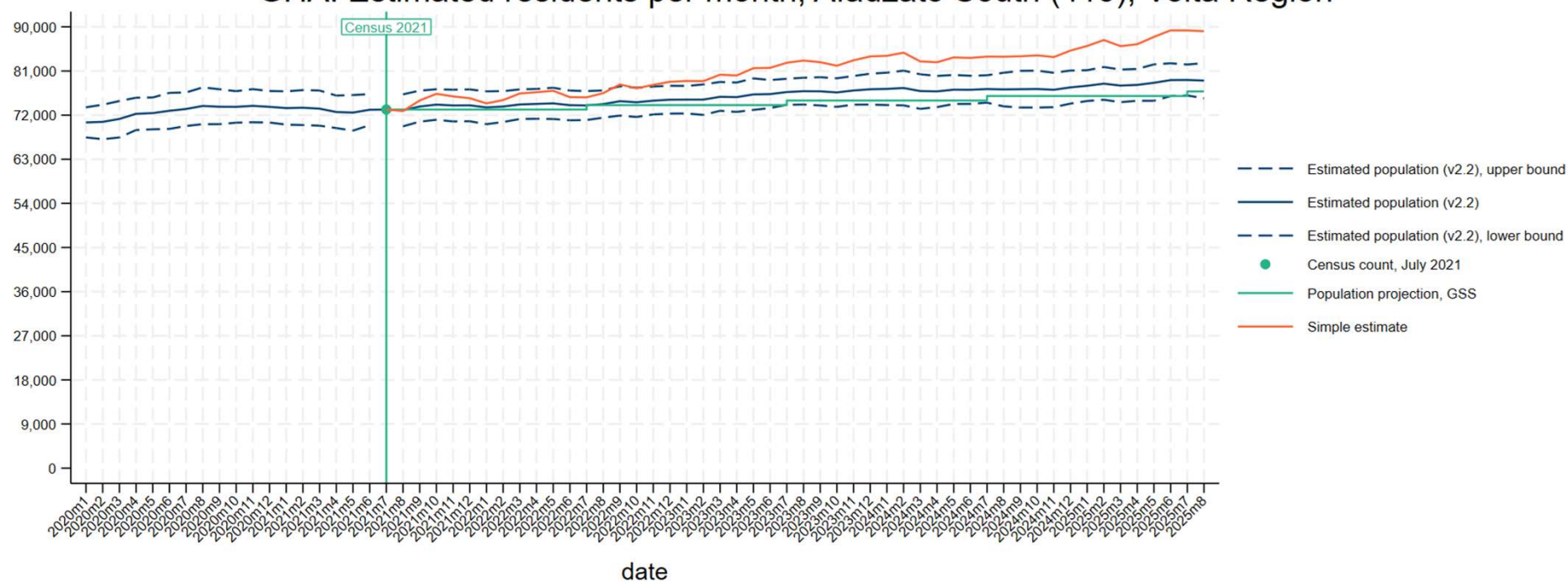
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Based on the information in the graphs, try to answer the following questions

- How does the GSS (annual) population projection compare to the (monthly) population estimates, based on demographic accounting? Which trends over time can be identified, and how do they differ between the data?
- How do the monthly estimates based on the **simple method** compare to the **more refined method**? Which estimates show larger or smaller variation over time?
- Based on the confidence intervals shown for the more refined estimates, which conclusions would you draw about the population changes in a given district, compared to the official GSS population projections?
- Based on the remaining limitations of estimation approaches discussed in the webinar, which caveats would you add to the interpretation?
- If you were to advise a policy maker in Ghana, how would you present these data, which conclusions would you draw, and which recommendations for using the data would you give?

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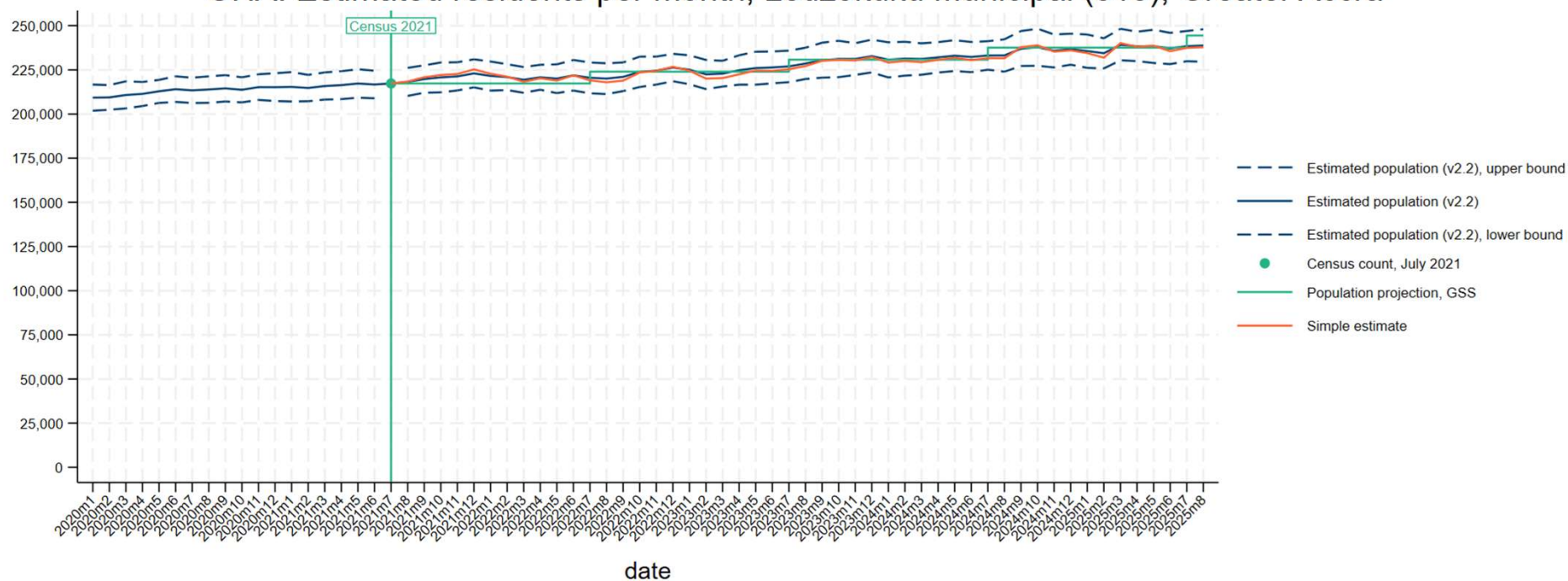
GHA: Estimated residents per month, Afadzato South (415), Volta Region



GHA estimates v2.2

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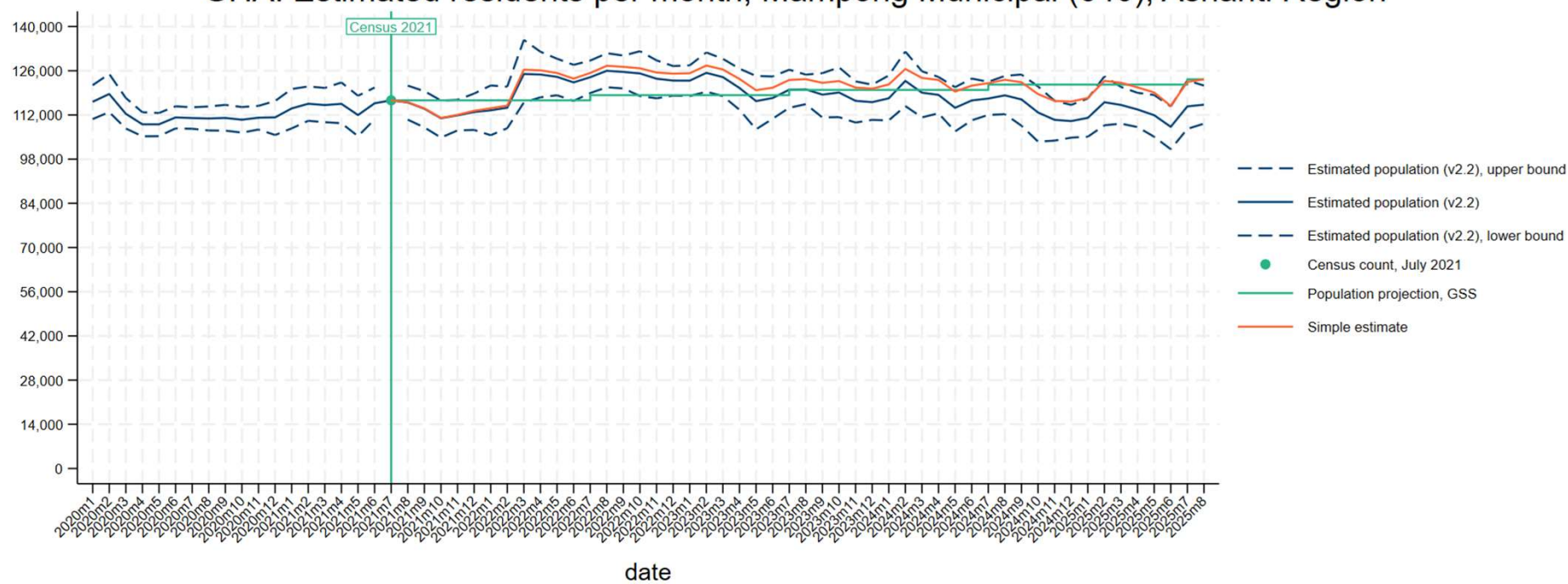
GHA: Estimated residents per month, Ledzokuku Municipal (313), Greater Accra



GHA estimates v2.2

Review activity

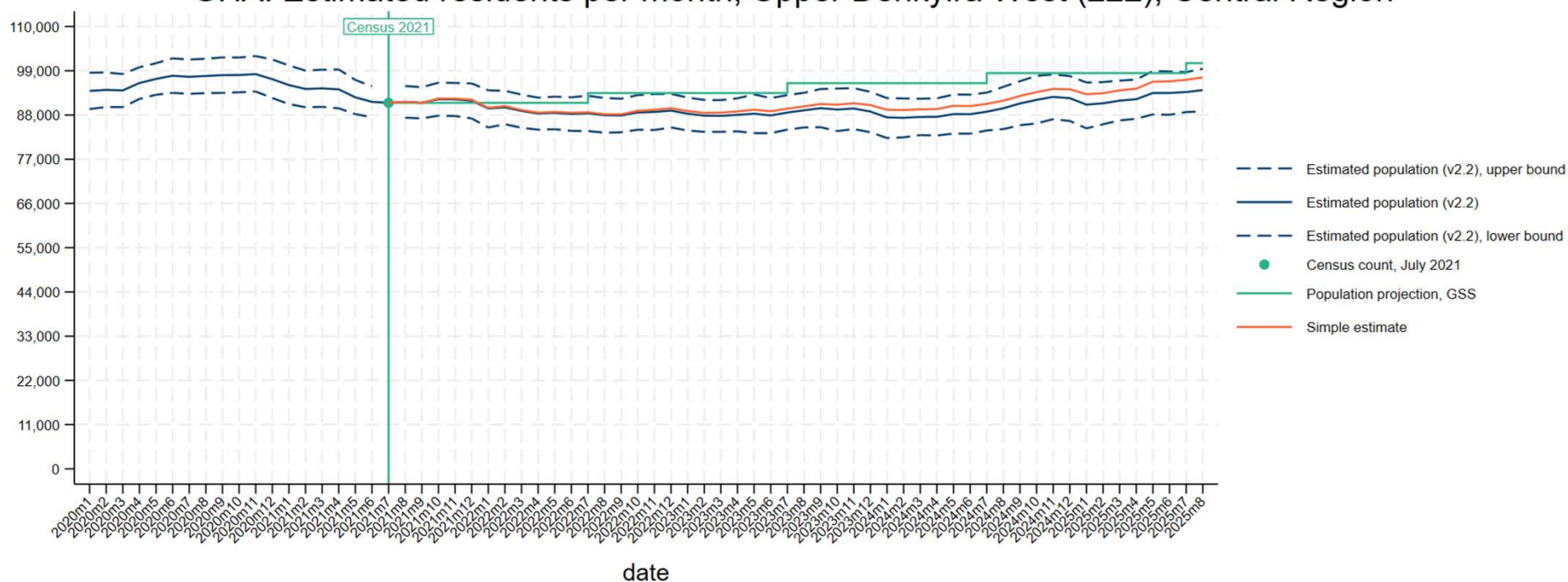
GHA: Estimated residents per month, Mampong Municipal (640), Ashanti Region



GHA estimates v2.2

Review activity

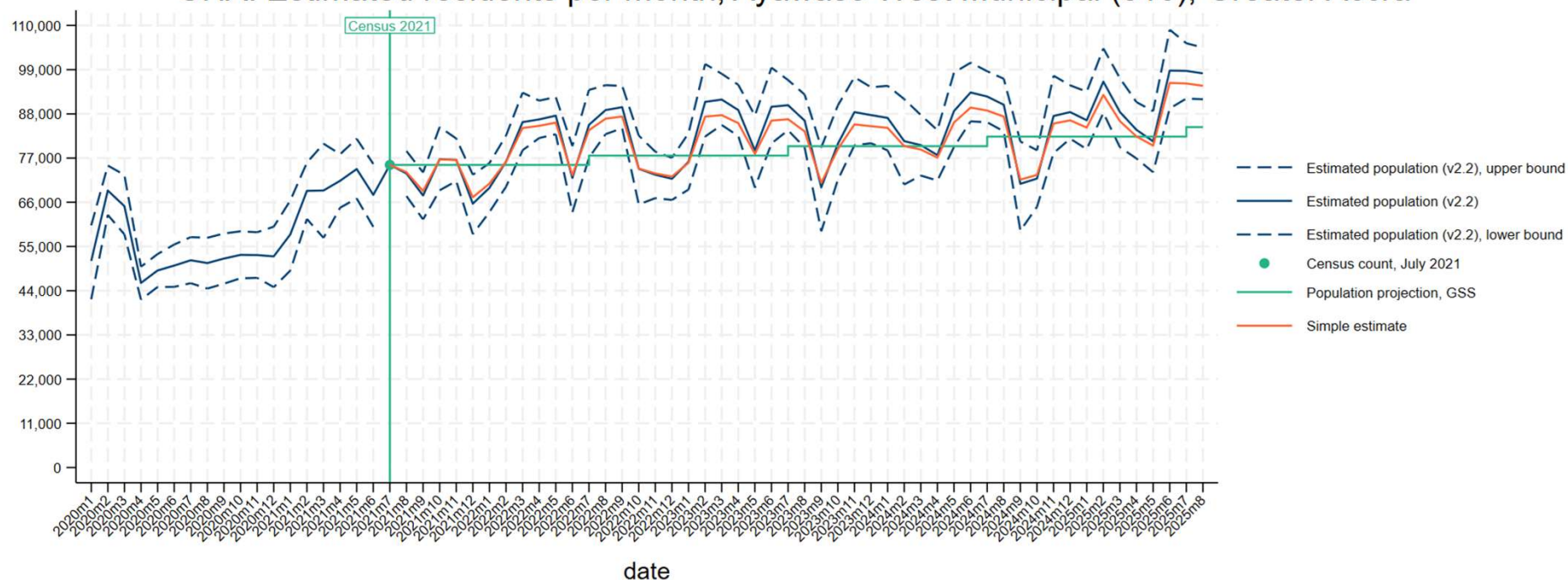
GHA: Estimated residents per month, Upper Denkyira West (222), Central Region



GHA estimates v2.2

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GHA: Estimated residents per month, Ayawaso West Municipal (316), Greater Accra



GHA estimates v2.2